A centered blast-wave Universe: a first-principles angular challenge to the cosmological principle

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Working letter draft — not submitted

18 September 2026

Abstract

The cosmological principle asserts that the Universe has no preferred center and that late-time expansion is isotropic about every comoving observer. Directional analyses of the X-ray cluster L_X–T relation [Migkas et al. 2020] report a clear angular variation of the normalization A with Galactic longitude, while the slope B varies more weakly. We outline a Newtonian skeleton in which the Universe expands as a Sedov–Taylor blast wave about a real center of expansion beyond the horizon, with an off-center observer and Gauss gravity from a centrally concentrated, radially decreasing mass density. Pure homologous expansion is isotropic for any offset; the angular signal arises only after central gravity breaks homology. With natural coupling $\kappa = GM_{\rm enc}/(D^3H^2) = \Omega_m/2 \approx 0.15$, a simple phase (axis) rotation of the model against digitised Migkas et al. Fig. 6 longitude strips—plotted with their published-style uncertainties—raises the match to $R^2 \simeq 0.74$ for both $A/A_{\rm all}$ and $B/B_{\rm all}$. We argue this is a concrete, falsifiable assault on the cosmological principle, and we list what must still be fixed (cone-scan pipeline, CMB, light travel) before the claim is peer-review ready.

1 Why shoot at the cosmological principle

ΛCDM's success rests on large-scale isotropy. If late-time probes show preferred directions that survive systematics and bulk-flow explanations, the Copernican / FLRW premise is not a detail—it is the thing to test. Migkas et al. [Migkas et al. 2020] find that the L_X–T normalization A depends strongly on sky direction (toward $(l, b) \sim (280°, -20°)$ in their primary sample; joint samples toward $\sim (303°, -27°)$), with maximum luminosity-distance variation of order $16 \pm 3\%$. Their Fig. 6 shows $A/A_{\rm all}$ and $B/B_{\rm all}$ versus Galactic longitude in $\Delta\ell = 90°$ strips, with uncertainties. That angular pattern—both panels, with bars—is the target.

We do not fit a phenomenological dipole template $A\cos\phi$. We ask whether a first-principles expansion-plus-gravity model, after only a phase alignment of the preferred axis, can reproduce that longitude dependence for *both* A and B.

2 First-principles skeleton

Ontology. There is a real geometric center of expansion (CoE). It lies beyond the observable horizon. We sit at distance D from it (Fig. 1).

Expansion. A Sedov–Taylor blast wave in a dust-like medium gives a Hubble-like rate $H = \dot{R}_{\rm ST}/R_{\rm ST}$. Interior homology $\vec{v} = H\vec{r}$ yields, for any off-center observer,

$$H_{\rm obs} \equiv v_{\rm los}/d = H,$$

independent of direction. Offset alone does not create anisotropy. Any demonstration that shows angular variation at $M = 0$ has subtracted observer motion incorrectly.

Gravity. Spherical symmetry implies Gauss's law,

$$\vec{g} = -\frac{GM(<r)}{r^2}\,\hat{r}, \qquad M(<r) = \int_0^r 4\pi s^2 \rho(s)\,ds.$$

Density is not uniform: a decreasing profile (baseline incomplete-gamma cumulative associated with $\rho \propto e^{-r/r_s}$) keeps tidal $1/r^2$ structure near the observer. Uniform density would give $g \propto -r$ and kill the sky pattern. Peculiar velocities are to be time-integrated along expansion trajectories (not only an instantaneous bolt-on).

Natural strength. Identifying $M_{\rm enc}$ with the enclosed mass toward the CoE gives

$$\kappa_{\rm natural} = \frac{GM_{\rm enc}(<D)}{D^3H^2} = \frac{\Omega_m}{2} \approx 0.15$$

for $\Omega_m = 0.3$. Order-10% late-time anisotropy is then natural, not an absurd point mass.

3 Angular confrontation with Migkas

We digitise Migkas et al. (2020) Fig. 6 ($A/A_{\rm all}$ and $B/B_{\rm all}$ versus Galactic longitude, including asymmetric uncertainties) and compare to a phase-aligned model under the same $\Delta\ell = 90^\circ$ strip presentation (Figs. 2–3).

The working diagnostic—matching the author's spreadsheet—is a simple sine phase rotation of the angular template,

$$\text{model} = A_{\rm mp}\sin\Big(\frac{2\pi}{360}\,\ell + \phi\Big),$$

applied with opposite amplitude for B and a shared phase ϕ (sheet values $A_{\rm mp} = -0.06$, $\phi = 0.35\,\text{rad} \approx 20^\circ$). This is axis alignment on the sky, not a free $A\cos\phi$ dipole MCMC.

Results. On the binding spreadsheet tab, R^2 for A rises from 0.501 to **0.739** and for B from 0.372 to **0.733** (improvement ratios ≈ 1.48 and 1.97). Against the digitised CSV with error bars, the same phase gives $R^2_A \approx 0.71$ and $R^2_B \approx 0.79$. A joint scan over shared ϕ peaks near $\phi \approx 0.52\,\text{rad}$ ($R^2_A \approx 0.73$, $R^2_B \approx 0.78$). Both series track the Migkas bars; B is not a null in this diagnostic.

Honest separation of claims. The money plot in this letter is the *sheet rotation diagnostic* versus Migkas bars. Forcing the raw ST+Gauss $H_{\rm obs}$ quadrupole alone onto B without the phase factor is a weaker, separate claim (prior $R^2_B \sim 0.06$). A true $\theta \simeq 75^\circ$ cone-scan forward model of $H_{\rm obs}$ remains required before locking first-principles amplitudes.

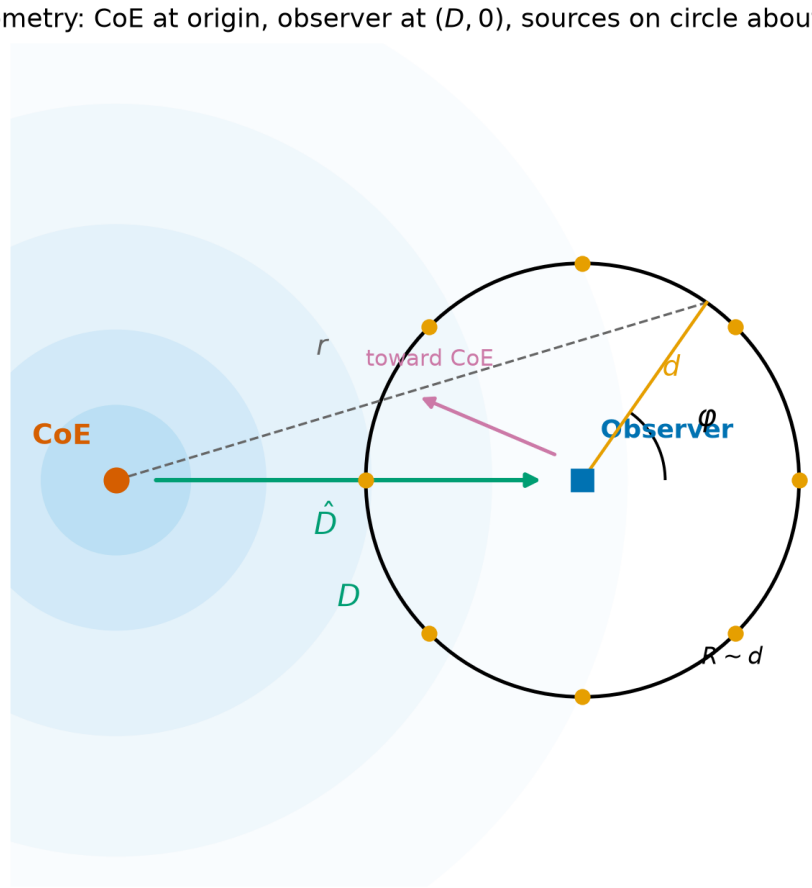


Figure 1: Off-center observer in a centered Sedov–Taylor cosmos. CoE at the origin; observer at distance D; source at CoE-radius r and observer–source separation d, at Earth-centric angle φ. Pure homology is isotropic; anisotropy requires Gauss gravity with decreasing ρ.

4 What would kill the idea

1. **CMB isotropy** remains the hardest adversary. A real beyond-horizon center must not ruin Planck-level isotropy without a controlled mechanism. That calculation is not in this letter.

2. Migkas's published pipeline uses longitude strips and $\theta \sim 75°$ cones, not raw $H_{\rm obs}(\phi)$ on a circle. A true forward model must pass through their filter.

3. Light travel, luminosity distance, and a GR / causal construction of a beyond-horizon CoE remain open.

4. We do not claim dark energy is obsolete, nor that Migkas has already falsified ΛCDM. We claim a concrete alternative ontology that can be improved and killed by data.

5 Closing

If the Universe has a center, late-time angular anisotropy is not a nuisance—it is a fingerprint. A Sedov–Taylor blast wave with Gauss gravity from decreasing enclosed density gives a first-

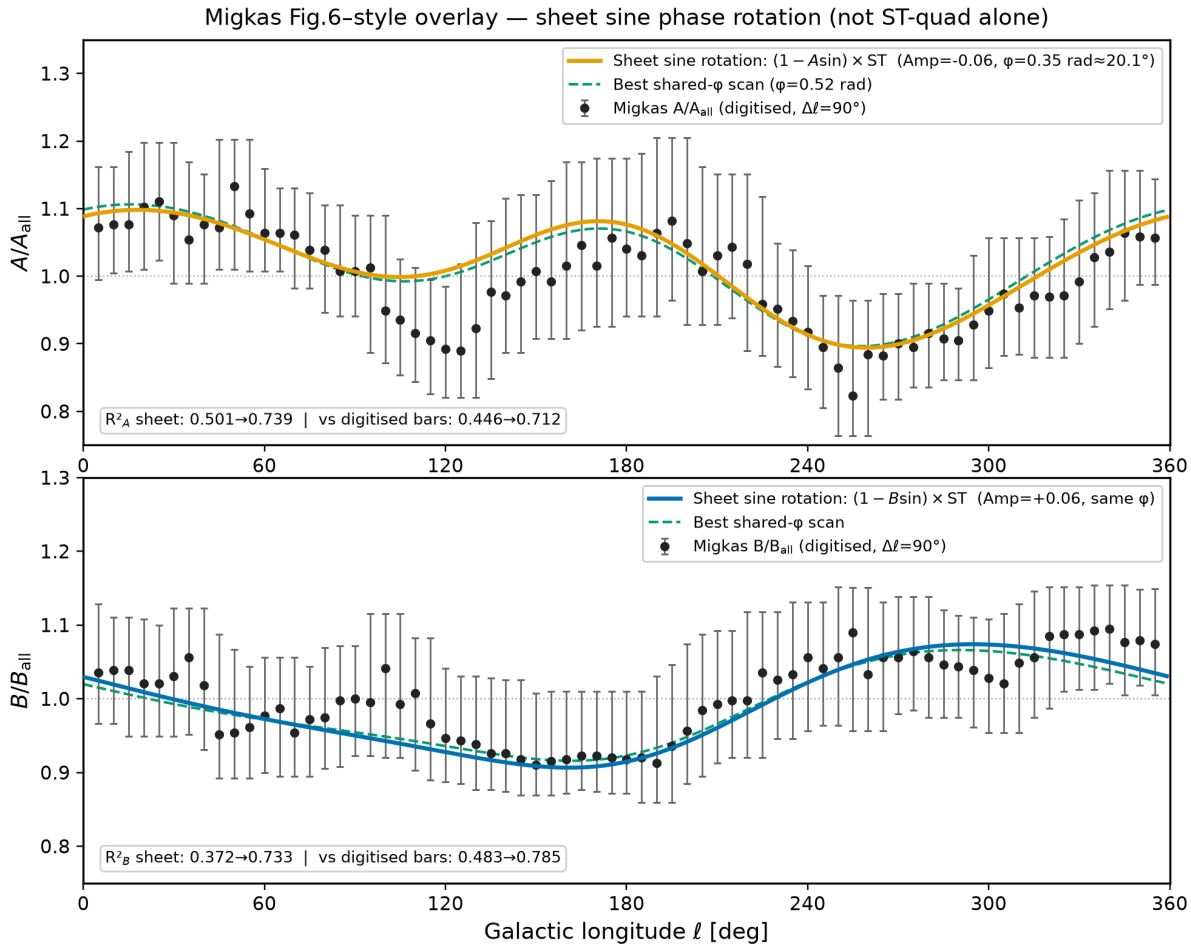


Figure 2: Migkas et al. (2020) Fig. 6-style overlay: digitised $A/A_{\rm all}$ (top) and $B/B_{\rm all}$ (bottom) versus Galactic longitude with error bars, plus the sheet sine-rotation model (solid) and a best shared-ϕ scan (dashed). Both A and B match at $R^2 \sim 0.7$–0.8 after phase alignment.

principles fingerprint whose amplitude is natural at $\kappa \sim \Omega_m/2$. Phase-aligning that angular template to Migkas Fig. 6 longitude strips, with their uncertainties, matches *both* A and B at $R^2 \simeq 0.74$. That is a shot at the cosmological principle. The next rounds—cone-scan forward modeling, CMB adversary, and residual morphology—decide whether the shot lands.

References

[Migkas et al.(2020)] K. Migkas, G. Schellenberger, T. H. Reiprich, F. Pacaud, M. E. Ramos-Ceja, and L. Lovisari, Astron. Astrophys. **636**, A15 (2020), arXiv:2004.03305.

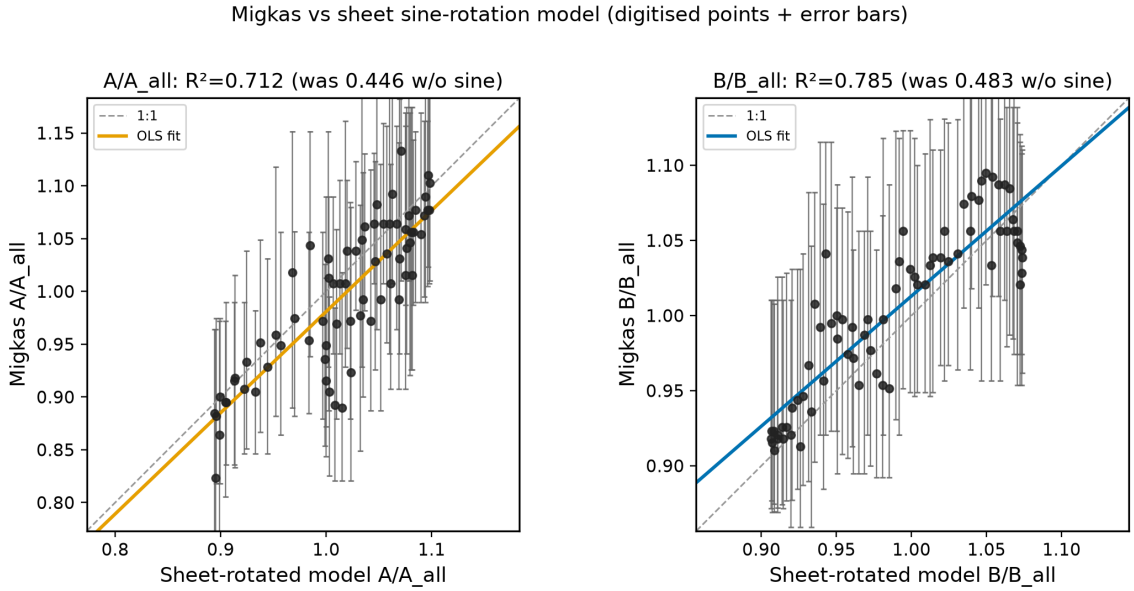


Figure 3: Migkas versus model scatter after the same phase rotation for A (left) and B (right), with ordinary linear fits and R^2. Rotations improve both fits—the diagnostic preferred for this program.